



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Batch: A-2

Division: A

Subject: Programming with Java

ASSIGNMENT No : 22

TITLE : Demonstration and exploration of JDBC classes and interfaces (PreparedStatement, and ResultSet).

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (PreparedStatement, and ResultSet).

Learning Objectives

To implement PreparedStatement and ResultSet for secure query execution and data retrieval.

Theory

PreparedStatement is an advanced JDBC interface used to execute precompiled SQL statements with parameters. Unlike the Statement interface, PreparedStatement allows SQL queries to be compiled only once and executed multiple times with different values, thereby improving performance. It also protects applications against SQL injection attacks by separating SQL code from user input. The ResultSet interface stores the records returned by SQL SELECT queries. It provides various methods such as next(), getInt(), getString(), and getDouble() for accessing retrieved data. PreparedStatement and ResultSet together

provide secure, efficient, and flexible database interaction and are widely used in enterprise Java applications.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp22/Main.java
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp22.Main
Enter Staff ID: DOC101
Enter Password: docpass
Login Successful!
Access Granted: Welcome, Doctor DOC101
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

CODE 1

```
1 package Exp22;
2
3 import java.sql.*;
4 import java.util.*;
5
6 public class Main {
7     public static void main(String[] args) {
8         Scanner sc = new Scanner(System.in);
9
10        System.out.print("Enter Staff ID: ");
11        String staffId = sc.nextLine();
12
13        System.out.print("Enter Password: ");
14        String password = sc.nextLine();
15
16        String query = "SELECT role FROM hospital_staff WHERE staff_id = ? AND
password = ?";
17
18        try (Connection con = DriverManager.getConnection(
19            "jdbc:mysql://localhost:3306/hospital_db",
20            "root",
21            "devansh@2007");
22            PreparedStatement ps = con.prepareStatement(query)) {
23
24            ps.setString(1, staffId);
25            ps.setString(2, password);
26
27            ResultSet rs = ps.executeQuery();
28
29            if (rs.next()) {
```

```

30         String role = rs.getString("role");
31         System.out.println("Login Successful!");
32
33         System.out.println("Access Granted: Welcome, " + role + " " + s
taffId);
34     } else {
35
36         System.out.println("Login Failed: Invalid Staff ID or Password.
");
37     }
38 } catch (Exception e) {
39     System.out.println("Database Error: " + e.getMessage());
40 }
41 sc.close();
42 }
43 }

```

EXERCISE 1

```

1 package Exp22.Exercise;
2
3 import java.sql.*;
4 import java.util.*;
5
6 public class Exp1 {
7     private static final String DB_URL = "jdbc:mysql://localhost:3306/
hospital_db";
8     private static final String DB_USER = "root";
9     private static final String DB_PASSWORD = "devansh@2007";
10
11     public static void main(String[] args) {
12         Scanner scanner = new Scanner(System.in);
13
14         System.out.println("=== Hospital Staff Portal ===");
15         System.out.print("Enter Staff ID: ");
16         String staffId = scanner.nextLine();
17
18         System.out.print("Enter Password: ");
19         String password = scanner.nextLine();
20
21         // Parameterized query for security and precompilation

```

```

22         String query = "SELECT role FROM hospital_staff WHERE staff_id = ? AND
password = ?";
23
24         try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_
PASSWORD));
25             PreparedStatement pstmt = conn.prepareStatement(query)) {
26
27             pstmt.setString(1, staffId);
28             pstmt.setString(2, password);
29
30             try (ResultSet rs = pstmt.executeQuery()) {
31                 if (rs.next()) {
32                     String role = rs.getString("role");
33                     System.out.println("\n Authentication Successful!");
34
35                     // Display role-specific access messages
36                     if ("Doctor".equalsIgnoreCase(role)) {
37
38                         System.out.println("Access Granted: Welcome, Dr. " + st
affId + ". Redirecting to Doctor Dashboard (Patient Records & Prescriptions).")
;
39                     } else if ("Nurse".equalsIgnoreCase(role)) {
40
41                         System.out.println("Access Granted: Welcome, Nurse " +
staffId + ". Redirecting to Nursing Station (Vitals & Ward Management).");
42                     } else {
43
44                         System.out.println("Access Granted: Welcome, Staff memb
er.");
45                     }
46                 } else {
47
48                     System.out.println("\n Authentication Failed: Invalid Staff
ID or Password.");
49                 }
50             } catch (Exception e) {
51                 System.err.println("Database Error: " + e.getMessage());
52             }
53         scanner.close();
54     }
55 }

```

EXERCISE 2

```
1 package Exp22.Exercise;
2
3 import java.sql.*;
4 import java.util.*;
5
6 public class Exp2 {
7     private static final String DB_URL = "jdbc:mysql://localhost:3306/
hospital_db";
8     private static final String DB_USER = "root";
9     private static final String DB_PASSWORD = "devansh@2007";
10
11     public static void main(String[] args) {
12         Scanner scanner = new Scanner(System.in);
13
14         System.out.println("=== System Login ===");
15         System.out.print("Enter Username: ");
16         String username = scanner.nextLine();
17
18         System.out.print("Enter Password: ");
19         String password = scanner.nextLine();
20
21         // SQL Query with placeholders (?) to prevent SQL Injection
22
23         String query = "SELECT * FROM users WHERE username = ? AND password = ?
";
24
25         try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_
PASSWORD);
26             PreparedStatement pstmt = conn.prepareStatement(query)) {
27             // Set parameters
28             pstmt.setString(1, username);
29             pstmt.setString(2, password);
30
31             try (ResultSet rs = pstmt.executeQuery()) {
32                 if (rs.next()) {
33                     System.out.println("\n Login Successful! Welcome, " + rs.ge
tString("username") + ".");
34                 } else {
```

```

35         System.out.println("\n Invalid Username or Password. Access
Denied.");
36     }
37 }
38
39     } catch (Exception e) {
40         System.err.println("Database Error: " + e.getMessage());
41     }
42     scanner.close();
43 }
44 }

```

SQL CODE:

```

1  -- Database: hospital_db
2  CREATE DATABASE hospital_db;
3  USE hospital_db;
4
5  -- Table for Exercise 1
6  CREATE TABLE users (
7      id INT PRIMARY KEY AUTO_INCREMENT,
8      username VARCHAR(50) NOT NULL,
9      password VARCHAR(50) NOT NULL
10 );
11 INSERT INTO users (username, password) VALUES ('admin', 'admin123');
12
13 -- Table for Exercise 2
14 CREATE TABLE hospital_staff (
15     id INT PRIMARY KEY AUTO_INCREMENT,
16     staff_id VARCHAR(50) UNIQUE NOT NULL,
17     password VARCHAR(50) NOT NULL,
18     role VARCHAR(20) NOT NULL -- 'Doctor' or 'Nurse'
19 );
20 INSERT INTO hospital_staff (staff_id, password, role) VALUES
21 ('DOC101', 'docpass', 'Doctor'),
22 ('NUR201', 'nurpass', 'Nurse');
23

```

Exercise

Q1. Create a login application using `PreparedStatement` to check username and password.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp22/Exercise/Exp1.java
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp22.Exercise.Exp1
=== Hospital Staff Portal ===
Enter Staff ID: DOC101
Enter Password: docpass

Authentication Successful!
Access Granted: Welcome, Dr. DOC101. Redirecting to Doctor Dashboard (Patient Records & Prescriptions).
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Create a Java application using `PreparedStatement` to authenticate hospital staff login (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp22/Exercise/Exp2.java
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp22.Exercise.Exp2
=== System Login ===
Enter Username: admin
Enter Password: admin123

Login Successful! Welcome, admin.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

Thus, the experiment successfully demonstrated the use of `PreparedStatement` and `ResultSet` in JDBC. `PreparedStatement` was used for secure and efficient execution of parameterized SQL queries, while `ResultSet` was used to retrieve and access records returned by `SELECT` queries. This provided an understanding of secure database interaction using JDBC.